

RESEARCH ARTICLE

An adaptive degradation-based maintenance model taking into account both imperfect adjustments and AGAN replacements

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Abstract

This paper presents an adaptive maintenance model for equipment that can be adjusted (minor preventive maintenance, imperfect state) or replaced (major preventive maintenance, as good as new) at specific scheduled times based on degradation measurements. An initial reliability law that uses a degradation-based model is built from the collection of hitting times of a failure threshold. Inspections are performed to update the reliability, the remaining useful life, and the optimum time for preventive maintenance. The case of both as good as new replacements and imperfect adjustments is considered. The proposed maintenance model is based on the optimization of the long-term expected cost per unit of time. The model is then tested on a numerical case study to assess its effectiveness. This results in an improvement for the occurrences of maintenance tasks that minimizes the mean cost per unit of time as well as an optimized number of adjustments that can be considered before replacing an item. The practical application is a decision aid support to answer the 2 following questions: Should we intervene now or wait for the next inspection? For each intervention, should we adjust or replace the item of equipment? The originality is the presence of 2 criteria that help the maintainer to decide to postpone or not the preventive replacement time depending on the measured degradation and to decide whether the item should be adjusted or replaced.

KEYWORDS

degradation-based reliability, imperfect maintenance, reliability-based maintenance, remaining useful life, stochastic processes

1 | INTRODUCTION

It is now well established that the maintenance sector is a keystone to ensure the competitiveness of any industry of productivity, quality, and availability. According to MIL-STD-3034, standard maintenance (preventive, corrective, and inactive) is the action of performing tasks (time-directed, condition-directed, failure-finding, servicing, and lubrication) at periodicities (periodic, situational, and unscheduled) to ensure that the item's functions (active, passive, evident, and hidden) are available until the next scheduled maintenance period. Generally speaking, both preventive

maintenance (PM) and corrective maintenance (CM) tasks are performed on industrial equipment through their operational lifetime. The balance between PM and CM actions usually depends on long-term cost rate, availability, or safety criteria. In this way, different maintenance strategies can be encountered in literature.^{1,2} They concern the replacement of systems subject to random failures and whose states are known at all time.

However, in some particular cases, the item state may be influenced by some factors especially for mechanical units that have to cope with variable mechanical stresses, a variable energy consumption, a modification of the

operating conditions, and the effect of the environment.³ Obviously, the reliability and remaining useful life (RUL) of such equipment will change accordingly. As a consequence, the maintenance scheduling has to be updated considering the degradation history. This topic is covered by the degradation-based reliability approach that consists in monitoring degradation covariates with respect to a given threshold to trigger inspection or maintenance events.

Finally, one also has to consider the efficiency of the maintenance action. Usually, the assumption of as good as new (AGAN) repairs is chosen, but there is a need to consider imperfect maintenance actions since item performances may decrease over time due to ageing or due to the impacts of consecutive adjustments.⁴

The objective of this paper is to extend an adaptive maintenance model to take into account both several consecutive imperfect adjustments and intermittent AGAN replacements.⁵ Degradation data are tracked to trigger the event times for maintenance tasks that consist in an inspection, an adjustment (imperfect PM), a PM replacement (AGAN), or a CM replacement. In fact, the degradation data are used to update a dynamical reliability model as well as the RUL. The paper is organized as follow:

- In the first part, we present the degradation-based reliability approach as well as a review of recent techniques for RUL estimation. Afterwards, we focus the study on the stochastic processes.
- The second part is dedicated to the adaptive maintenance model. This model is built on both preventive and corrective actions to optimize the long-term cost rate. We also present an extension of this model that takes into account the inspection costs. Two heuristic criteria are proposed to build this model.
- The third part summarizes the methodology that allows obtaining a dynamic specific reliability and the predictive time for replacement from new degradation data input that are obtained during the inspection of a unit. The maintenance decision framework is also presented.
- The fourth part shows the results of the methodology on an illustrative example subdivided in 3 cases depending on the shape of the degradation trend using a nonstationary gamma process.

2 | DEGRADATION-BASED RELIABILITY AND RUL ESTIMATION

2.1 | Notations and definition

Let $Z(t)$ be the evolution of the degradation over time, z_f (a positive value) be the failure threshold, and $X(t_j)$ be a degradation measurement at inspection time t_j . We suppose that

the degradation process leads to a soft failure, which means that there are no other hard failure modes that compete for the failure time. Let T be the failure time defined by⁶

$$T = \inf \{t : Z(t) \geq z_f\}. \quad (1)$$

The reliability $R(t)$ is then the probability for the degradation process to remain under the given threshold:

$$R(t) = P(T > t) = P(Z(t) < z_f). \quad (2)$$

Under the concept of first hitting time of a given threshold, the RUL of an item given the conditional measurement $X(t_j)$ at inspection time t_j and the preset threshold z_f is

$$\text{RUL}(t_j) = \inf \{l : X(t_j + l) \geq z_f | l \geq 0, X(t_j) < z_f\}. \quad (3)$$

To obtain an accurate estimation of the RUL, the degradation model $Z(t)$ should perfectly fit the degradation data $X(t)$ to minimize the error in the forecasted degradation value $Z(t_j + l)$. Practically, the RUL is obtained by the computation of the mean residual life (MRL) (ie, the mean value of the RUL conditioned by the observations $X(t_j)$) using the following equation:⁷

$$\begin{aligned} \text{MRL}(t_j) &= E(T - t | T > t, X(t_j) < z_f) \\ &= \int_t^\infty R(u | X(t_j)) du, \end{aligned} \quad (4)$$

where $R(u | X(t_j))$ denotes the conditional reliability of the item at time $u > t_j$, given the degradation measurement $X(t_j)$.

2.2 | Degradation models

Remaining useful life knowledge is a keystone to offer guidance for sequential management involving inspection schedule, maintenance, replacement, and spare parts ordering. It has been considered as a foundational ingredient in the field of prognostics and health management.⁸ The RUL estimation can be achieved through a degradation-based modeling of the item state. According to the literature, there are mainly 3 approaches for RUL estimation that are the physical model-based methods, the data-driven methods, and their combination.⁹ While physical model-based methods (also known as physics-of-failure models) allow to deeply explain the degradation mechanisms that lead to the failure, it is rather difficult to adapt this approach on an operating system due to the diversity and uncertainty of its environment. Meanwhile, data-driven methods have become an interesting alternative to evaluate both the dynamic reliability and the RUL. An exhaustive review was achieved by Si et al¹⁰ on data-driven methods for both direct and indirect observed degradation data with a presentation of the relevant methods such as the use of gamma processes, Wiener processes, Markov models, stochastic regression models, stochastic filtering-based

models, and covariate hazard-based models to list the merits and drawbacks. The main challenge of RUL estimation lies in the presence of heterogeneity due to different inner states or external operating conditions of systems. Indeed according to Zhang et al.,⁸ the performance degradation of a system is caused by interactions of both the inner deterioration and the working environment of the system, justifying the need to take into account the heterogeneity in the degradation model. In this way, the authors identified 3 kinds of heterogeneity that are the unit-to-unit variability for items from the same batch, the variability in the operating conditions over time, and the diversity of tasks and workloads of systems during their life cycles. For each heterogeneity, the authors recommended the most adequate degradation models. We briefly present those models in the following subsections with a focus on the unit-to-unit variability and time-varying dynamics of systems.

2.2.1 | Random coefficient regression models

Random effects first appeared in random coefficient regression models.¹¹ They are used when the degradation is represented by a set of discrete values $\{X_{ij}\}$. At each inspection time t_j , a degradation value $X(t_j)$ is measured on an item of equipment i . The degradation model takes the form

$$X_{ij} = Z(t_{ij}; \alpha; \beta_i) + \varepsilon_{ij}, \quad (5)$$

with $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)$ is a vector of constant parameters that are characteristics of the tested population, and $\beta_i = (\beta_{i1}, \beta_{i2}, \dots, \beta_{in})$ is a vector of random parameters that are specific to each item of equipment i (ie, α is the vector representing the common part of the degradation while β_i represents the heterogeneity). The term ε_{ij} represents the measurement error on the degradation value at time t_j on the item i and is supposed to follow a Gaussian distribution with a null mean and a standard deviation σ_e . Common representations of this model are the linear, the power, and the logarithmic forms. As an application of the linear form, Freitas et al.¹² represented the evolution of tire wear. Gebræel et al.¹³ used this model to predict the RUL of bearings from a rotating machinery using vibration data. However, this simple model has several drawbacks including the need for more historical degradation data from different systems of the same category, the difficulty in capturing the time-varying dynamics of the systems, and the independency between random noise with time.¹⁴

2.2.2 | Stochastic process-based models

Stochastic process-based models with random coefficients are able to consider both time-varying dynamics and unit-to-unit variability. These processes may be represented by some specific models that derived from the Levy processes family.¹⁵ A Levy stochastic process has independent (non)stationary

increments, which represent the sequence of successive random and independent displacements of a point in a space. Common models from this family are the Wiener and the gamma process. The jumps of the Wiener process follow a normal distribution with mean m and standard deviation $\sigma\sqrt{t}$. The evolution of this process has a linear trend with local positive or negative jumps.¹⁶ On the opposite, the gamma process presents a monotonous evolution made of increments that follow a gamma distribution.¹⁷ Among the recent contributions in the study of stochastic processes for RUL estimation, Lawless and Crowder¹⁸ used a gamma stochastic process with covariates and random effects for degradation modelling and dynamic RUL computation. The case of nonmonotonic degradation processes with fluctuations was considered by Wang¹⁹ who used a Wiener degradation model with random effects. A degradation path-based RUL estimation method with the exact closed-form solution of the estimated power density function of the RUL in linear and exponential cases was proposed by Si et al.¹⁶ According to the results presented in the literature, it seems that stochastic process models with random effects can effectively improve the accuracy of RUL estimation in addition to extend the range of applications by considering both cases of monotonous and nonmonotonous degradation processes, whether they are linear or nonlinear.⁸ However, in industrial applications, the main drawback of stochastic process models with random effects is the computation issue that can be complex and highly dependent of the choice of random parameters and their distribution. Generally speaking, the assumption of normally distributed parameters is chosen.²⁰ The next section is dedicated to the study of the gamma process.

2.3 | The gamma process

In 1975, the gamma process was introduced in the area of reliability to model stochastic degradation and to optimize maintenance. Because the gamma processes are well suited for modeling the temporal variability of deterioration, they have proven to be useful in determining optimal inspection and maintenance decisions.²¹

In this paper, we use a generic gamma degradation process defined as follow:

$$Z(t) = G(m(t), \beta), \quad (6)$$

with $m(t)$ is the shape function and β is the scale parameter. A gamma process has the following properties:

- G has independent and stationary jumps (the second assumption is only valid for $m(t) = \alpha t$);
- $G(t=0) = 0$; and
- G is a continuous stochastic process; for $\forall t, h \geq 0$, the jump $G(t+h) - G(t)$ follows a gamma distribution with

shape parameter $m(h)$ and scale parameter β with density function:

$$f(x) = \frac{\beta^{m(h)}}{\Gamma(m(h))} x^{m(h)-1} \exp(-\beta x), \quad (7)$$

with Γ the gamma function define as follow $\Gamma(m(h)) = \int_0^{+\infty} x^{m(h)-1} \exp(-x) dx$.

In the following part, we consider a shape function $m(t) = at^p$. This choice allows to represent the 3 general trends for degradation processes, ie, concave ($p < 1$), linear ($p = 1$), and convex ($p > 1$).²¹ As the gamma function is defined in $\mathbb{R}^+$, this stochastic process can only produce positive increments leading to a monotonous degradation path that is well suited to represent physical degradation mechanisms such as wear, creep, or crack growth. One should notice that this process presents a time dependency (ie, the longer the elapsed time between sequential jumps, the higher their intensity). This implies that the gamma process may only be used in applications for which the degradation jumps are proportional to the elapsed time or the number of usages.

The expected value and the variance of a gamma process are, respectively,²¹

$$E(G(t)) = \frac{m(t)}{\beta} = \frac{at^p}{\beta}, \quad (8)$$

$$V(G(t)) = \frac{m(t)}{\beta^2} = \frac{at^p}{\beta^2}. \quad (9)$$

In this study, we use the nonstationary gamma process for simulation purposes mainly. Obviously, the parameters of the gamma process may be estimated from measured degradation data. Methods to estimate the gamma process are the method of maximum likelihood, moments, Bayesian statistics, and the method of expert judgement that are exhaustively discussed by Van Noortwijk.²¹ Finally, the analytical reliability of a gamma degradation process reaching a given threshold z_f is given by the following equation:⁶

$$\begin{aligned} R(t|z_f, m(t), \beta) &= 1 - \frac{\Gamma(m(t), \beta(z_f - z_0))}{\Gamma(m(t))} \\ &= 1 - \frac{\int_{\beta(z_f - z_0)}^{+\infty} x^{m(t)-1} \exp(-x) dx}{\Gamma(m(t))} \\ &= \frac{\int_0^{\beta(z_f - z_0)} x^{m(t)-1} \exp(-x) dx}{\int_0^{+\infty} x^{m(h)-1} \exp(-x) dx}. \end{aligned} \quad (10)$$

However, this analytical reliability may lead to computational issue, and one should prefer to compute the distribution of the threshold hitting times through a classical

statistical analysis based on generated degradation paths by Monte Carlo simulations.²²⁻²⁴

3 | MAINTENANCE MODEL

3.1 | General assumptions

- The proposed approach is applicable for equipment that can be adjusted or replaced.
- The failure time of an item of equipment is ruled by a stochastic degradation process, ie, it corresponds to the hitting time of a degradation threshold.
- Whenever a failure occurred, a CM action is conducted immediately, ie, the degradation process cannot cross the threshold and there is no duration in failed state to consider.
- The degradation process itself is not altered by the maintenance actions, ie, the parameters of the degradation process remain unchanged. Maintenance actions only affect the recovery values of the degradation at inspection times.
- Both PM and CM are considered, ie, the failure of an item of equipment does not lead to dramatic consequences (only higher cost values). An optimal inspection time T_p is obtained considering the balance between preventive and corrective costs.
- At each inspection time, 2 strategies are considered depending on the measured degradation level. If the degradation is lower than expected, then the maintenance action is postponed to another inspection time considering the future degradation pattern. Otherwise, the maintenance action is conducted at current inspection time. The decision to do or postpone the maintenance action is given by the first criterion K_C .
- A PM action is either an adjustment (imperfect maintenance) or a replacement (perfect AGAN maintenance), the choice resulting from the assessment of the second criterion K_m .
- In case of an adjustment, the degradation process is set to a value z_0 that is lower than the degradation value $z(t_j)$ measured at inspection time t_j . In this study, it is considered that the intensity of the recovery value decreases in proportion with the number of adjustments that were already conducted on the item of equipment (the higher the number of adjustments, the lesser the efficiency).
- The value z_0 obtained after the adjustment action is measured so that the recovery efficiency is also known by the maintenance staff, ie, it will conditioned the next inspection time given the fact that the MRL of the item of equipment can be assessed.
- In case of a CM or a preventive replacement, the item of equipment is replaced by a new one (perfect AGAN maintenance) so that $z_0 = 0$, ie, the expected life of the

item of equipment corresponds to its mean time to failure (MTTF).

- The adaptive maintenance model with imperfect maintenance actions rely on 2 criteria: the first one will assess the optimal time for a PM actions considering the balance between preventive replacement cost and corrective replacement cost given an updated degradation-based reliability model. The second criterion will recommend the best PM action to conduct (adjust or replace) given the cost to residual life expectancy ratio and the balance between the adjustment cost and preventive replacement cost. Actually, the risk of a failure is assessed in the first criterion; the second criterion supposes that the item of equipment has survived until the scheduled inspection time.
- It should be highlighted that for the assessment of the second criterion K_m , the maintenance team firstly proceed to the adjustment of the item of equipment in all cases. The improvement in the degradation level z_0 is then measured so that the updated MUT can be computed and the K_m is assessed. In case of finally opting for a replacement ($K_m < 1$), the cost of the adjustment is not incurred, ie, it is assumed that this cost is counterbalanced by the residual value of the item of equipment that is removed so that in the end, only the cost of the AGAN preventive replacement is due.
- Downtimes due to unavailability of the maintenance staff or resources are not considered at this stage, ie, the duration of maintenance actions are considered as negligible compared to the life duration of the asset.

3.2 | Notations and definition

The proposed maintenance model is based on the reliability calculation that allows to obtain the most relevant time to achieve the PM. Although this approach is old and well known, we propose to use it in the context of a reliability model that is updated for each new degradation measurement or failure time. We recall the economic model proposed by Lyonnet²⁵ that distinguishes the losses due to a stop (P_{arr}) and the costs due to an intervention (C_{int}).

The losses (P_{arr}) are composed of

- production losses $P_{prod} = \tau_{prod} T_{arr}$, with T_{arr} being the duration of a stop and τ_{prod} the loss rate per unit of time;
- raw material losses P_{mp} ;
- amortization of breakdown equipment P_{am} ; and
- energy consumption P_e .

The intervention costs (C_{int}) are the sum of following costs:

- salary costs $C_s = \tau_s T_{int}$, with T_{int} the elapsed time to achieve the intervention and τ_s the salary per unit of time;
- storage costs for failed items C_o ;
- amortization of maintenance equipment $C_{am} = \tau_{am} T_{int}$, with τ_{am} the amortization rate per unit of time; and
- spare items costs C_r .

Both losses and costs are made of a constant contribution—subscript “cst”—and a contribution that is proportional to the elapsed time. For the following developments, we denote by the subscript “arr” the variable losses per unit of time due to a stop and by the subscript “int” the variable costs per unit of time due to an intervention to obtain

$$P_{arr} = \tau_{arr} T_{arr} + P_{cst}, \quad (11)$$

$$C_{int} = \tau_{int} T_{int} + C_{cst}. \quad (12)$$

To simplify the analytical cost assessment Lyonnet²⁵ stated that the stop duration (T_{arr}) and the elapsed time for an intervention (T_{int}) are equal. Next, we use the subscript “c” to specify the parameters related to a CM and the subscript “p” to identify the parameters of a PM. The time for preventive replacement is optimized to reduce the overall expected maintenance cost per unit of time. The next parts present the theoretical calculations to find these optima. We also define the following parameters:

- T_p is the time of PM;
- $F(T_p)$ represents the probability of having a failure at time T_p given the degradation-based reliability model;
- C_c is the corrective cost when a failure occurs;
- C_p is the cost when a PM is achieved;
- $\bar{c}_m$ is the average cost per unit of time that has to be optimized;
- $MUT|T_p$ is the mean up time under a PM policy;
- $MTTR_c$ and $MTTR_p$ are the mean time to restore respectively for a CM and for a PM; and
- MTTF the mean time to failure

3.3 | Cost optimum

Generally speaking, the mean cost per unit of time $\bar{c}_m$ can be divided into CM costs C_c and PM costs C_p . To calculate these different costs, some models have been detailed in literature.²⁵ Mean maintenance costs can be determined in function of the systematic time of PM and form this relationship:

$$\bar{c}_m(T_p) = \frac{1}{\text{MUT}|T_p} \{F(T_p)(P_{\text{arr}_c} + C_{\text{int}_c}) + [1-F(T_p)](P_{\text{arr}_p} + C_{\text{int}_p})\}, \quad (13)$$

with

$$P_{\text{arr}_c} = \text{MTTR}_c \tau_{\text{arr}} + P_{\text{cst}_c}, \quad (14)$$

$$C_{\text{int}_c} = \text{MTTR}_c \tau_{\text{int}_c} + C_{\text{cst}_c}, \quad (15)$$

$$P_{\text{arr}_p} = \text{MTTR}_p \tau_{\text{arr}} + P_{\text{cst}_p}, \quad (16)$$

$$P_{\text{arr}_p} = \text{MTTR}_p \tau_{\text{int}_p} + C_{\text{cst}_p}, \quad (17)$$

$$\begin{aligned} \bar{c}_m(T_p) = \frac{1}{\text{MUT}|T_p} \{ & F(T_p)[\text{MTTR}_c(\tau_{\text{arr}} + \tau_{\text{int}_c}) + C_{\text{cst}_c} \\ & + P_{\text{cst}_c}] + [1-F(T_p)][\text{MTTR}_p(\tau_{\text{arr}} + \tau_{\text{int}_p}) \\ & + C_{\text{cst}_p} + P_{\text{cst}_p}]\}, \end{aligned} \quad (18)$$

$$\bar{c}_m(T_p) = \frac{F(T_p) \cdot C_c + R(T_p) \cdot C_p}{\text{MUT}|T_p} = \frac{F(T_p) \cdot C_c + R(T_p) \cdot C_p}{\int_0^{T_p} R(t) dt}, \quad (19)$$

with C_c and C_p are the overall maintenance costs for corrective and preventive actions, respectively. As for a pure CM policy ($T_p = \infty$), Equation 19 becomes

$$\frac{\bar{c}_{mc}}{\int_0^\infty R(t) dt} = \frac{\text{MTTR}_c(\tau_{\text{arr}} + \tau_{\text{int}_c}) + C_{\text{cst}_c} + P_{\text{cst}_c}}{\text{MTTF}}. \quad (20)$$

3.4 | Inspection costs

To update the degradation-based reliability model, the item of equipment often has to be shut down to allow the inspection. This may lead to an extra cost due to nonproduction and man hours as well as a decrease of the availability. According to the literature survey, there is no reference model that describes the contributions of the inspections costs. Generally speaking, papers that deal with the optimization of inspection costs refer to a generic cost model of the form²⁶⁻²⁹

$$C(t) = C_i N_i(t) + C_p N_p(t) + C_c N_c(t) + C_d d(t), \quad (21)$$

with

- C_i is the global cost of an inspection;
- $N_i(t)$ the number of inspections achieved within the time interval $[0, t]$;
- C_p the global cost of a PM intervention;
- $N_p(t)$ the number of PM actions achieved within the time interval $[0, t]$;
- C_c the global cost of a CM intervention;
- $N_c(t)$ the number of CM actions achieved within the time interval $[0, t]$;
- C_d the unavailability cost rate per unit of time; and
- $d(t)$ is the cumulative elapsed time of unavailability recorded within the time interval $[0, t]$.

Using the same formalism as the one presented in section 3.1 with the subscript “ i ” to denote an inspection, we introduce the inspection costs in Equation 13:

$$\begin{aligned} \bar{c}_m(T_p^*) = \frac{1}{\text{MUT}|T_p^*} \{ & F(T_p^*)(P_{\text{arr}_c} + C_{\text{int}_c}) \\ & + [1-F(T_p^*)](P_{\text{arr}_p} + C_{\text{int}_p}) + N_i(T_p^*)(P_{\text{arr}_i} + C_{\text{int}_i}) \}, \end{aligned} \quad (22)$$

with

$$P_{\text{arr}_i} = \text{MTTI} \tau_{\text{arr}} + P_{\text{cst}_i}, \quad (23)$$

$$C_{\text{int}_i} = \text{MTTI} \tau_{\text{int}_i} + C_{\text{cst}_i}, \quad (24)$$

MTTI being the mean time to conduct an inspection. It follows

$$\begin{aligned} \bar{c}_m(T_p^*) = \frac{1}{\text{MUT}|T_p^*} \{ & F(T_p^*)[\text{MTTR}_c(\tau_{\text{arr}} + \tau_{\text{int}_c}) + C_{\text{cst}_c} \\ & + P_{\text{cst}_c}] + [1-F(T_p^*)][\text{MTTR}_p(\tau_{\text{arr}} + \tau_{\text{int}_p}) + C_{\text{cst}_p} \\ & + P_{\text{cst}_p}] + N_i(T_p^*)[\text{MTTI}(\tau_{\text{arr}} + \tau_{\text{int}_i}) + C_{\text{cst}_i} + P_{\text{cst}_i}] \}, \end{aligned} \quad (25)$$

$$\bar{c}_m(T_p^*) = \frac{F(T_p^*) \cdot C_c + R(T_p^*) \cdot C_p + N_i(T_p^*) \cdot C_i}{\int_0^{T_p^*} R(t) dt}, \quad (26)$$

with C_i the global inspection costs. The notation T_p^* indicates that the schedule of the PM may be different than its initial value T_p (ie, the planning will be updated given the degradation assessment during the inspection). The problem consists in determining the optimum number of inspections that allows the item of equipment to run as long as possible

but without being too costly due to these inspections. We introduce a heuristic cost criterion $K_C(t_j)$ that helps to decide to postpone or not the initial time for a preventive replacement given the degradation level measured during the inspection:

$$K_C(t_j) = \frac{\frac{(j-1) \cdot C_i + C_p}{t_j}}{\frac{F(T_p^*) \cdot C_c + R(T_p^*) \cdot C_p + j \cdot C_i}{\int_0^{T_p^*} R(t) dt}}, \quad (27)$$

with $T_p^* = T_p(t_j|Z(t_j))$ the next forecasted time for a preventive replacement given the degradation level $Z(t_j)$ measured at the j th inspection time t_j . This criterion represents the ratio between the strategy that consists to replace the item of equipment at time t_j and the strategy to postpone the replacement at time T_p^* . Actually, the numerator represents the cost rate obtained for a lifecycle t_j and the denominator the cost rate for an expected lifecycle T_p^* that is obtained given the last degradation measurement and considering the cost values (see Section 4). While $K_C > 1$, it is cheaper to postpone the PM action to the next T_p^* calculated. When $K_C \leq 1$, we reached the optimal number of inspections N_i and the maintenance is achieved at the last inspection time. A strong assumption is required; however, at each inspection time, the maintenance staff is ready for a PM if necessary.

Instead of considering events that occurred since time 0 for the item of equipment, one may consider the elapsed time between inspection times to consider the effect of an additive inspection cost, the cost criterion becomes

$$K_C(\Delta t_j) = \frac{\frac{C_p}{\Delta t_j}}{\frac{F(T_p^*) \cdot C_c + R(T_p^*) \cdot C_p + C_i}{\int_{t_{j-1}}^{t_j} R(t) dt + \int_{t_j}^{T_p^*} R(t) dt}}, \quad (28)$$

with $\Delta t_j = t_j - t_{j-1}$ the time interval between 2 consecutive inspections. At inspection time t_j and if the item of equipment has not failed, $\int_{t_{j-1}}^{t_j} R(t) dt = \Delta t_j$. The cost equation may then be computed by

$$K_C(\Delta t_j) = \frac{C_p}{F(T_p^*) \cdot C_c + R(T_p^*) \cdot C_p + C_i} \cdot \left(1 + \frac{\int_{t_j}^{T_p^*} R(t) dt}{\Delta t_j} \right). \quad (29)$$

3.5 | Imperfect maintenance actions

The concept of imperfect maintenance actions was already introduced in the late 70s.¹ Actually, after a PM, the unit is usually younger but sometimes, it might be worse than before because of faulty procedures, eg, wrong adjustments, bad

parts, and damage occurring during the maintenance. Generally, the improvement of the unit by PM would depend on the resources spent for PM.³⁰ Pham and Wang⁴ summarized the treatment methods to take into account imperfect maintenance actions. Nakagawa first proposed to treat the imperfect maintenance as follow: the unit has a probability p to be restored in AGAN state and a probability $q = 1 - p$ to be in as bad as old (ABAO) state. It follows that if $p = 1$, then the maintenance is perfect and if $p = 0$, then the maintenance is inefficient. This simple (p, q) rule allows to consider a general imperfect maintenance model for which AGAN and ABAO maintenance are special cases. Block et al³¹ extended the (p, q) by taking into account the age of the unit at maintenance time thus obtaining a time varying $(p(t), q(t))$ rule. Other imperfect maintenance models focus on the failure rate value after PM. If the failure rate is unchanged, then maintenance is ABAO; if the failure rate is restored to its initial value, then maintenance is AGAN; the degree of improvement in the failure rate is called the improvement factor. Also, considering the failure rate, Kijima et al³² introduced the notion of virtual age process of a repairable system and proposed 2 models for imperfect maintenance. Kijima model 1 assumes that the n th repair can remove the damage incurred between the $(n-1)$ th and the n th failures with a given maintenance efficiency while Kijima model 2 supposes that the last repair will remove the cumulative damage from both current and previous failures. The core idea of the virtual age models is to substitute the real time with the virtual age for failure intensity computation. Last but not least, imperfect maintenance model may also be considered for continuous/shock degradation processes. Kijima and Nakagawa³³ presented a cumulative damage shock model with imperfect periodic PM. The PM actions consist in the reduction of the damage level according to an efficiency parameter b , ie, each PM action reduces the cumulative damage by $100(1 - b)\%$. Obviously, this approach is very similar to the (p, q) rule; the difference resides in the fact that maintenance efficiency directly affects the degradation process itself rather than the equipment survival law or failure rate.

In this study, we consider imperfect maintenance actions that directly affect the degradation value. During a PM action, an item of equipment may be adjusted (minor imperfect PM) or replaced (major AGAN PM). After inspection, the cost for a minor maintenance is C_{p_minor} and the cost for a replacement is C_{p_major} (with $C_{p_minor} < C_{p_major}$). We make the assumption that each time a minor maintenance is achieved, the degradation value is set to a random value z_0 that is proportional to the cumulative number of minor adjustments already achieved. The question that arises when deciding to perform the PM, is whether the item of equipment should be adjusted or replaced? To answer that question, we propose the following second heuristic criterion:

$$K_m(x) = \frac{\frac{C_{p_major}}{MUT_{AGAN}|T_p}}{\frac{C_{p_minor}}{MUT_{notAGAN}^*(x)|T_p^*}}. \quad (30)$$

This criterion compares the cost to residual life expectation ratio ($C_{p_major}/MUT_{AGAN}|T_p$) due to an AGAN replacement under PM at T_p and the cost to residual life expectation ratio due to a x^{th} adjustment ($C_{p_minor}/MUT_{notAGAN}^*(x)|T_p^*$) under updated PM at time T_p^* given the degradation level. The updated mean up time $MUT_{notAGAN}^*(x)|T_p^*$ being a function of x is decreasing as the item is under imperfect maintenance; the cost to residual life expectation ratio is increasing for the imperfect PM case. In consequence while $K_m > 1$, it is cheaper to conduct a minor PM by adjusting the item and partially reducing its degradation level. And as soon as $K_m \leq 1$, the item of equipment has to be replaced by a new one.

4 | METHODOLOGY

In this section, we present the methodology that consists in updating a degradation-based reliability model from data as well as the maintenance optimization for preventive replacement. First, we start from an initial guess for the preventive time of replacement T_p . This can be obtained from a failure data analysis on identical equipment or from the recommendation of the constructor. This generic time of replacement must be quite safe in regard to the consequences of a failure. After that, one has to measure an indicator of the degradation for an item of equipment. According to its evolution, a degradation model $Z(t)$ may be chosen and the parameters of the model are estimated. From this model, degradation paths are simulated until they reach the failure threshold. The collection of the hitting times of the threshold defines a

new failure function. A new reliability model is then fitted on this failure function from which we can deduce the MRL as well as a new optimized time T_p^* for a preventive replacement. Figure 1 shows an illustration of the updating process of both the degradation and the threshold hitting times distribution.

We denote by $'^*$ ' any value or parametric law that is updated given the last degradation measurement achieved. As the reliability model $R^*(t)$ is updated each time new degradation data are collected, the maintenance model based on cost is also updated, respectively. This methodology permits to increase the useful life of an item of equipment for which the specific degradation path is quite optimistic compared to the mean trend and to obtain a better estimation of the MRL in general.

Figure 2 presents the flowchart of the adaptive maintenance model. The first step is the identification of the degradation model that best fits the degradation measurements. Goodness of fit criterion such as the determination factor for a regression or the maximum likelihood estimators computation are used to identify the most adequate law. In this study, we suppose that the degradation process $Z(t)$ is known and is not modified by maintenance actions. Once the degradation is identified, the second step consists in the simulation of random degradation paths and the computation of the hitting times of the degradation threshold corresponding to the failed state. From this collection of hitting times, a statistical generic reliability law is computed that allows to determine the first inspection time $t_j = T_p$ given the costs of the different maintenance actions. At this time and considering that the item has not failed, the inspection allows to obtain the current value of the degradation. From this value and given the degradation process, new degradation trajectories are simulated to obtain a new set

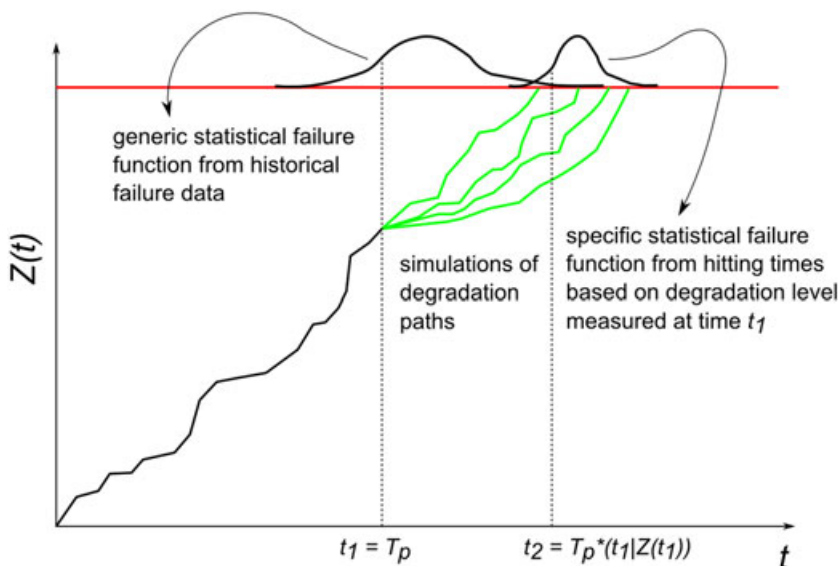


FIGURE 1 Illustration of the updated degradation-based reliability model. The knowledge of the degradation level allows obtaining an item-specific reliability model instead of the generic reliability model obtained from failure data [Colour figure can be viewed at wileyonlinelibrary.com]

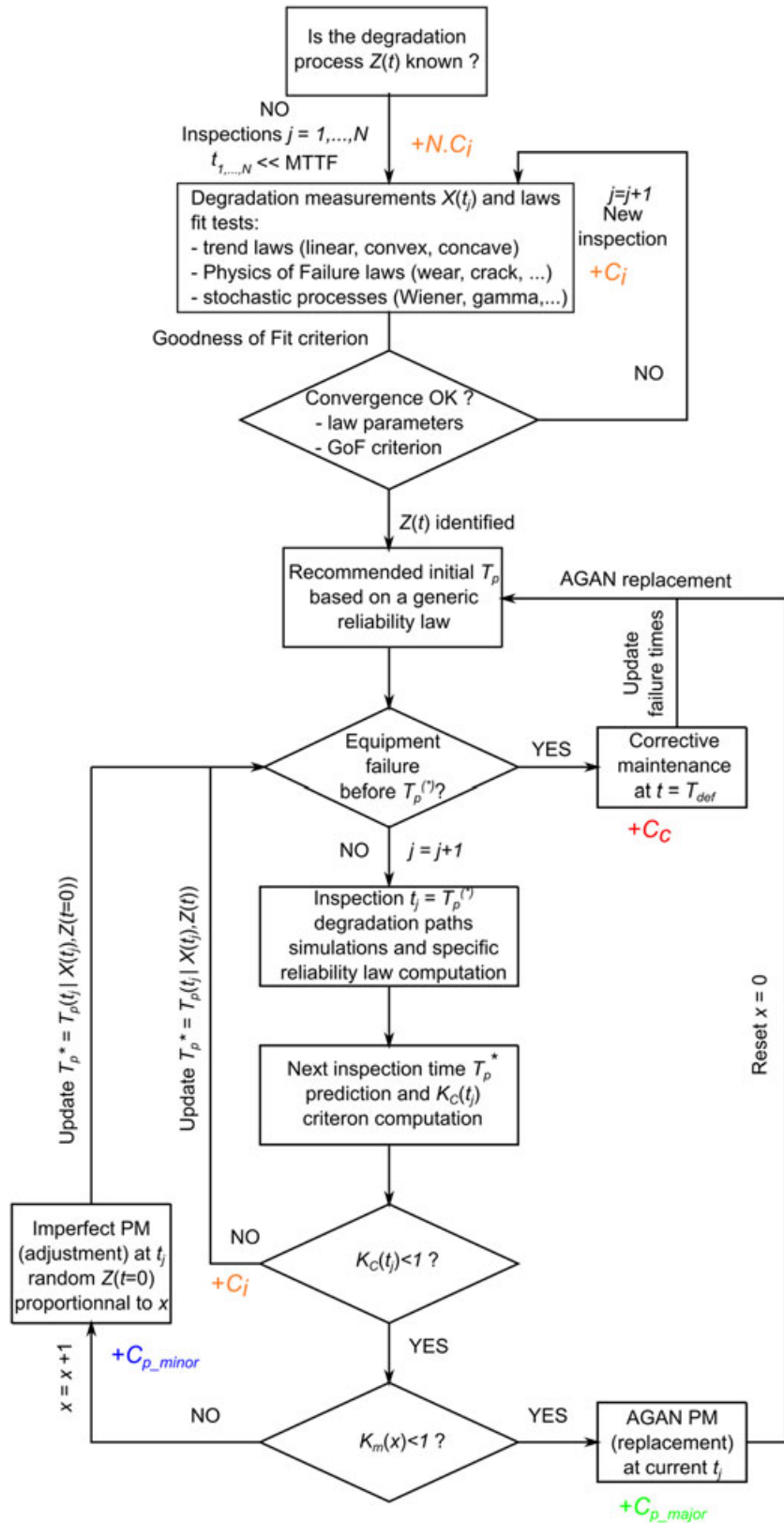


FIGURE 2 Methodology flowchart of the adaptive maintenance and associated maintenance costs. AGAN, as good as new; PM, preventive maintenance [Colour figure can be viewed at wileyonlinelibrary.com]

of hitting times of the threshold and the reliability law is updated accordingly. The third step is the assessment of the cost criterion $K_C(t_j)$ and the maintenance type criterion $K_m(x)$. The first cost criterion is used to decide whether or not a maintenance action should occur while the second

criterion specifies the maintenance action to be realized, ie, imperfect PM (adjustment) or AGAN PM (replacement). Figure 3 depicts the graphical interpretation of these criteria. Obviously, the item may fail between 2 inspection times that leads to a CM (AGAN replacement) and the time of failure

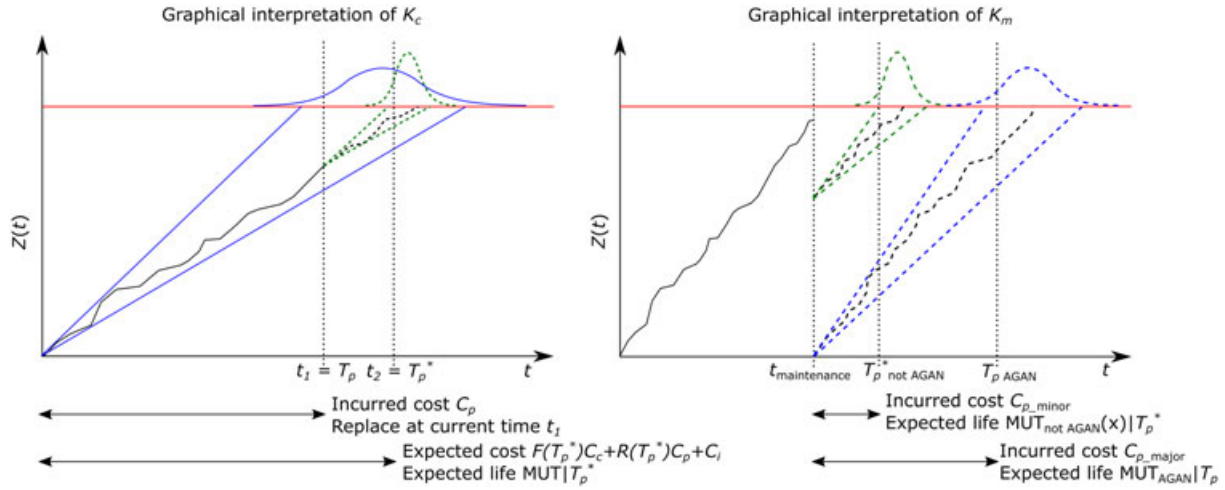


FIGURE 3 Graphical interpretation of the K_c and K_m criteria. On the left figure, deciding to postpone the maintenance action (ie, $K_c > 1$) allows to extend the useful life of the asset but will also increase the expected cost due to the contribution of C_i and the probability of a failure before next inspection. On the right figure, opting for an adjustment (ie, $K_m > 1$) leads to a partial improvement of the asset condition at the price of a minor cost C_{p_minor} . As the efficiency of these adjustments decreases, an as-good-as-new replacement with cost C_{p_major} is profitable after a given number x of adjustments [Colour figure can be viewed at wileyonlinelibrary.com]

is added on the failure times database to update the time of the first inspection.

5 | A NUMERICAL CASE STUDY

5.1 | Degradation model

In this part, we apply the methodology on pseudo degradation data obtained from a nonstationary gamma degradation process:

$$Z(t) = G(\alpha t^p, \beta). \quad (31)$$

In Table 1, we consider 3 different cases that correspond to a concave, a linear, and a convex law.

The degradation threshold value z_f that corresponds to failure is set to unity. The expected hitting time of the threshold for a new item is given by Equation 8:

$$E(Z(t)) = \frac{\alpha t^p}{\beta} = 1, \quad (32)$$

which leads to a MTTF that is 100 time units [TU] for both the 3 cases.

TABLE 1 Parameters of the nonstationary gamma process for 3 cases

Degradation Model	1. Concave	2. Linear	3. Convex
α	5	0.5	0.05
β	50	50	50
p	0.5	1	1.5

Figure 4 shows a random degradation trajectory as well as the expected value and the standard deviation band with respect to time for the 3 cases of the nonstationary gamma process.

For each case, we simulated 1000 degradation paths and the hitting times of the degradation threshold were computed. From this collection of hitting times, we fitted a lognormal distribution using a regression method and the median rank estimator.²² Figure 5 shows the generic reliability law of an AGAN item for the 3 cases. One can see that the concave degradation case leads to a higher dispersion than the 2 others cases despite having the same MTTF.

5.2 | Maintenance model

For the sake of simplicity, global costs are considered in this numerical case study. Obviously, for a real case study, the contributions of those generic costs should be detailed as presented in Section 3. We consider the following costs:

- $C_i = 100$ €
- $C_{p_minor} = 400$ €
- $C_{p_major} = 800$ €
- $C_c = 1200$ €

Table 2 shows the parameters of the lognormal distributions that were fitted by regression on the 3 cases as well as the optimized systematic replacement maintenance periodicity T_p and expected systematic preventive $\bar{c}_m(T_p)$ and corrective $\bar{c}_{mc}$ maintenance costs obtained by Equations 19 and 20. For the concave case, one can see that the maintenance periodicity T_p is very close to the MTTF

FIGURE 4 Simulations of the nonstationary gamma process (staircase style), expected value (dash style), and standard deviation interval (dot style) with respect to time for the 3 cases (red, concave law; blue, linear law; green, convex law) [Colour figure can be viewed at wileyonlinelibrary.com]

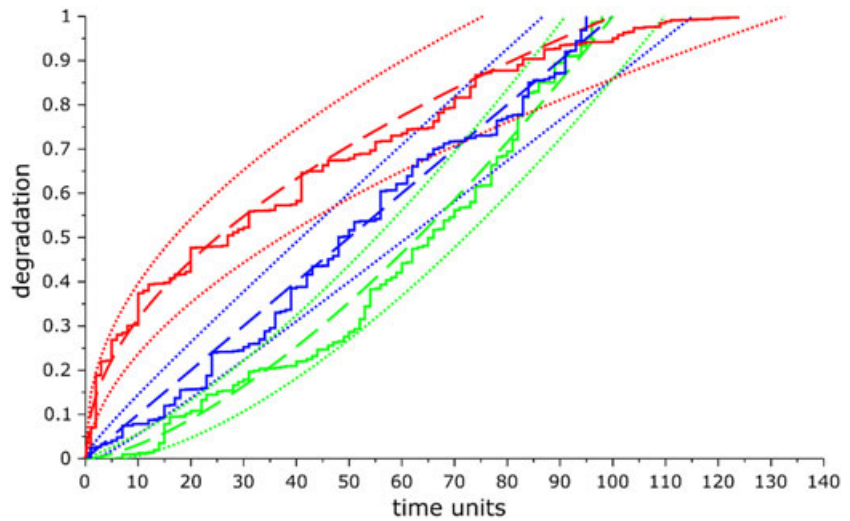


FIGURE 5 Initial reliability models (ie, from as-good-as-new state) for the 3 cases of the nonstationary gamma process. The concave case presents a higher dispersion than the 2 other cases [Colour figure can be viewed at wileyonlinelibrary.com]

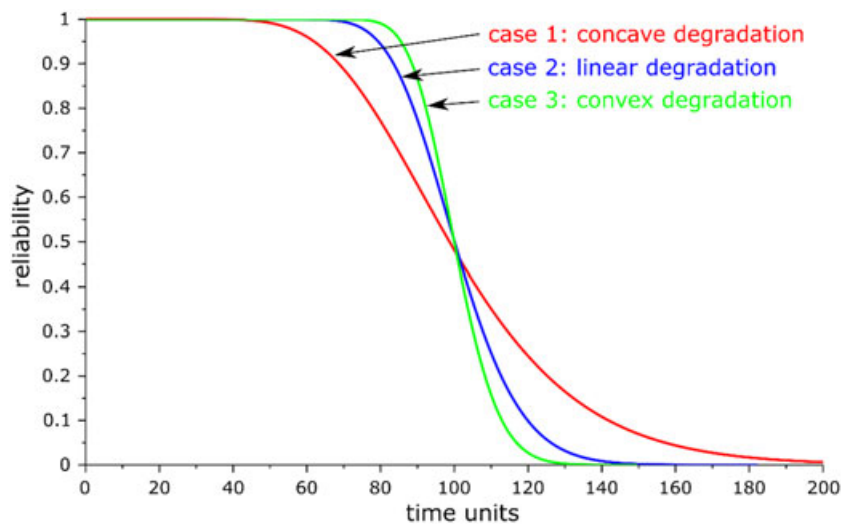


TABLE 2 Lognormal reliability parameters of AGAN item, optimized systematic maintenance periodicity, systematic preventive replacement, and corrective maintenance costs for the 3 cases

Law Characteristics	1. Concave	2. Linear	3. Convex
Lognormal μ [TU]	4.592	4.606	4.605
Lognormal σ [TU]	0.283	0.141	0.095
T_p optimum [TU]	98	87.8	88.7
$\bar{c}_m(T_p)$ [€/TU]	11.20	10.05	9.53
$\bar{c}_{mc}$ [€/TU]	12	12	12

(100 TU). This is due to the higher dispersion observed for this case and the balance between the preventive and corrective costs. In consequence, there are more uncertainties for the concave degradation case that lead to a higher systematic PM cost than the 2 other cases.

Whenever a preventive adjustment is achieved (imperfect maintenance), the degradation process is set to a value

z_0 that follows a uniform distribution with boundaries $[0.1x - 0.05; 0.1x + 0.05]$ with x being the number of adjustments (including the current one) already done for an item of equipment. If the item of equipment has to be replaced (AGAN maintenance) or has failed, the degradation is set to 0 and x is also reinitialized to 0. Figure 6 shows a comparison of the non-AGAN and AGAN strategies for 5 consecutive simulations of the gamma degradation process (convex case). One can see the optimization problem that consists in finding the optimal number of imperfect adjustments before replacing an item.

5.3 | Simulations

Figure 7 shows a snapshot of the simulation of the adaptive maintenance model considering inspections, non-AGAN adjustments, and AGAN replacements for the nonstationary gamma process (convex case). The crossed dots represent the event times that lead to maintenance actions. The color

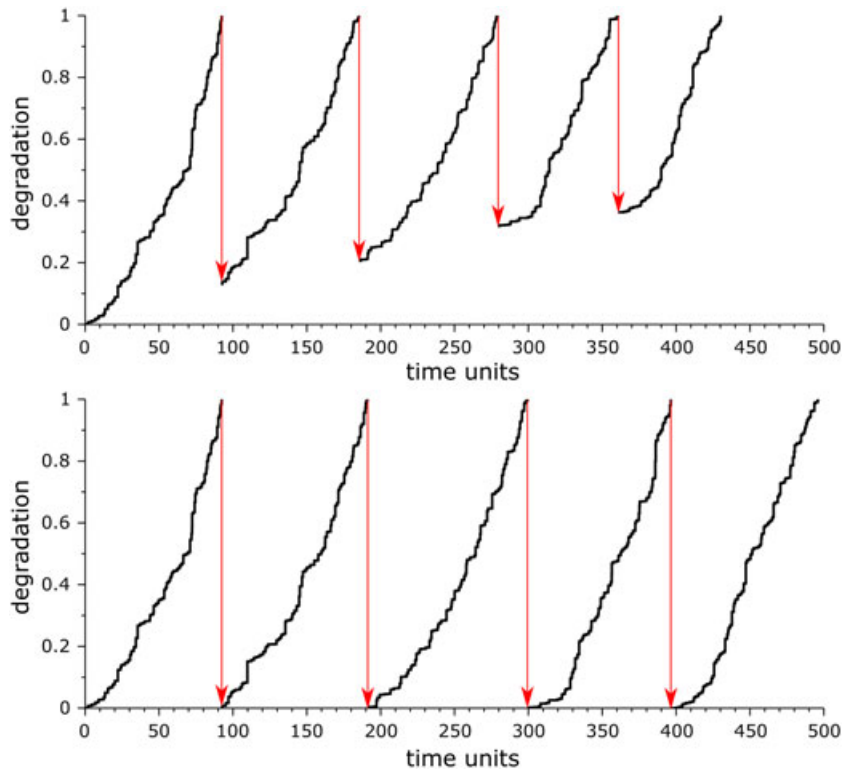


FIGURE 6 Comparison of the imperfect preventive maintenance and as-good-as-new maintenance [Colour figure can be viewed at wileyonlinelibrary.com]

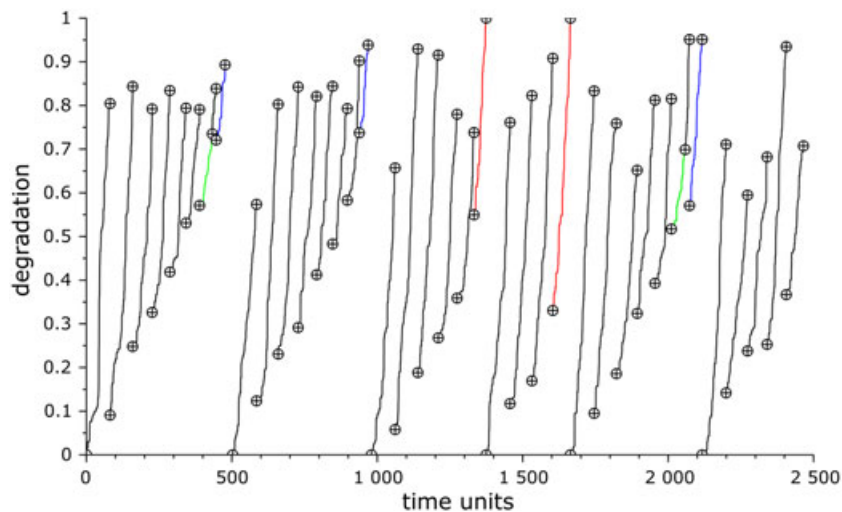


FIGURE 7 Snapshot of the simulations of the adaptive maintenance with imperfect adjustments (in black) and AGAN replacements (in blue). Inspections that resulted in no maintenance actions are represented in green and corrective replacements after failure are in red [Colour figure can be viewed at wileyonlinelibrary.com]

code on the degradation paths represents the nature of the maintenance action, ie, in black, the item was adjusted (imperfect PM); in blue, the item was replaced (AGAN PM); in green, an inspection occurred, but no maintenance action was required, and in red, a CM occurred after failure. For each case, 1000 simulations were sufficient to obtain a representative asymptotic cost per unit of time. As an example for the convex case on the 1000 runs, we recorded 34 CM actions due to failure, 110 major preventive replacements, 856 minor preventive adjustments, and 27 inspections resulting in no maintenance actions thus allowing a life extension of the unit.

5.4 | Results

Figures 8, 9, and 10 show the performance of this adaptive maintenance model with imperfect adjustments compared to the adaptive maintenance model with systematic AGAN replacements. Expected costs per unit of time for the systematic PM model (Equation 19) and pure CM model (Equation 20) are also represented as referenced values. For both the 3 degradation cases, the adaptive maintenance model considering both imperfect adjustment and AGAN replacements allows to decrease the expected maintenance cost per unit of time significantly compared to the adaptive

FIGURE 8 Expected maintenance cost for different maintenance policies. Concave degradation case. AGAN, as good as new [Colour figure can be viewed at wileyonlinelibrary.com]

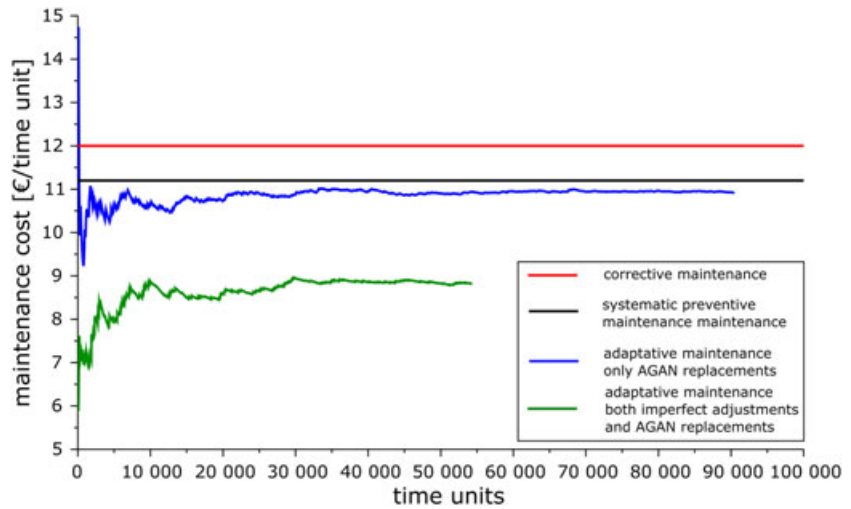


FIGURE 9 Expected maintenance cost for different maintenance policies. Linear degradation case. AGAN, as good as new [Colour figure can be viewed at wileyonlinelibrary.com]

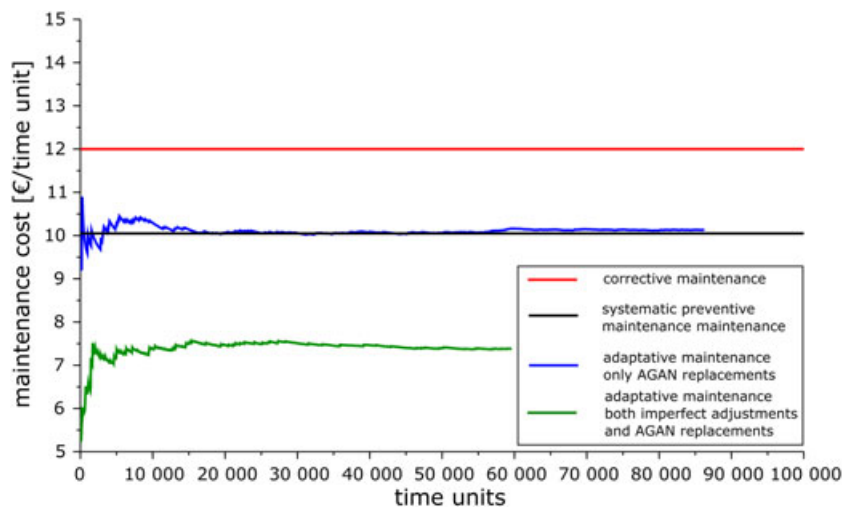
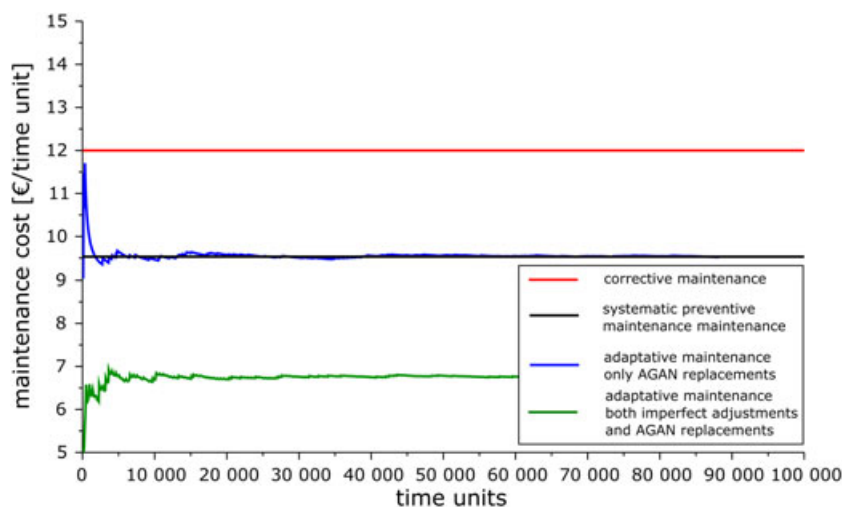


FIGURE 10 Expected maintenance cost for different maintenance policies. Convex degradation case. AGAN, as good as new [Colour figure can be viewed at wileyonlinelibrary.com]



maintenance model with systematic AGAN replacements (eg, -18.5% for the concave case, -26.9% for the linear case, and -28.6% for the convex case). The lowest maintenance cost value is reached for the convex case followed by the

linear case. This is due to the reliability law linked with the convex degradation-based model that presents the smallest dispersion, thus reducing the negative impact of randomness during the simulations.

TABLE 3 Influence of the minor preventive maintenance cost on the mean cost per unit of time for the 3 scenarios

Mean cost per unit of time [€]	Degradation model		
	1. Concave	2. Linear	3. Convex
$C_{p_minor} = 100 \text{ €}$	7.69	4.59	3.63
$C_{p_minor} = 200 \text{ €}$	8.04	5.86	5.01
$C_{p_minor} = 400 \text{ €}$	8.81	7.38	6.78
$C_{p_minor} = 600 \text{ €}$	9.56	7.75	7.47
$C_{p_minor} = 800 \text{ €}$	10.81	10.04	9.49

5.5 | Sensitivity analysis on C_{p_minor}

An originality of this study concerns the introduction of minor imperfect maintenance actions to decrease the degradation level of a unit. To assess the impact of the different cost ratios, a sensitivity analysis was conducted on the minor maintenance cost C_{p_minor} . Under the condition $C_i \leq C_{p_minor} \leq C_{p_major} \leq C_c$, 5 values were considered, ie, 100 €, 200 €, 400 €, 600 €, and 800 € while the other parameters remained unchanged. For these values, 1000 new simulations were achieved on each case, the results are presented in Table 3.

From the mean cost values obtained in Table 3, one can see that the relative influence of the minor PM cost is more significant for the scenarios 2 and 3 than for the scenario 1. As already mentioned for the concave scenario, the threshold hitting times prediction presents more uncertainties due to the shape of the reliability law and thus lead to more corrective actions, which explained the higher cost values obtained.

6 | CONCLUSION

In this paper, an adaptive maintenance model was proposed to consider various maintenance actions that are made of inspections, imperfect adjustments, AGAN preventive replacements, and corrective replacements. The event times for maintenance actions are conditioned by the degradation assessment of a unit health. In this way, the degradation-based reliability approach was presented with a focus on stochastic processes and RUL estimation. After that, the maintenance cost model was built considering the different costs contributions. Then we detailed the methodology that consists in updating the reliability and RUL given the last degradation measurements and predictive hitting times of the degradation process. A first originality consisted in the introduction of inspection cost that is required to assess the degradation value. As a second originality, 2 heuristic criteria were defined to construct the adaptive maintenance decision flowchart. The first criterion justifies whether a maintenance

action is needed or not while the second one helps to decide whether the equipment should be adjusted (imperfect PM) or replaced (AGAN PM). An illustrative example involving a nonstationary gamma process declined in 3 cases was studied to demonstrate the benefits of the advocated adaptive maintenance model. The results showed a reduction of the expected maintenance cost per unit of time for all the cases in particularly for the convex degradation case. Future work will consist in defining other forms of imperfect maintenance such as the modification of the degradation rate parameter or the shape parameter in the model. Moreover, there is still an open research area in the field of degradation process estimation from real data to relax the assumption that the degradation process is known a priori.

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